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MADRAS INSTITUTE OF TECHNOLOGY CAMPUS

DEPARTMENT OF COMPUTER TECHNOLOGY

II YEAR

ACADEMIC YEAR 2026 – 2027

DIGITAL COLLEGE LIBARY

Submitted by

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Under the Guidance of

Dr. Ponsy R.K. Sathia Bhama

CERTIFICATE

This is to certify that the project report entitled "DIGITAL COLLEGE LIBARY" is the bonafide work of RAMYA K (Register No: 2025503042), II Year, Department of Computer Technology, who carried out the project work under my supervision during the academic year 2026–2027.

Submitted for the project viva-voce examination held on _____.

Signature of Guide

Dr. Ponsy R.K. Sathia Bhama

Signature of Head of Department

Dr. Ponsy R.K. Sathia Bhama

DECLARATION

I, RAMYA K (Register No: 2025503042), hereby declare that the project report titled "DIGITAL COLLEGE LIBARY" is an original record of the work carried out by me under the guidance of Dr. Ponsy R.K. Sathia Bhama, Department of Computer Technology, and that this work has not been submitted, either in part or in full, for the award of any other degree or diploma of this or any other institution.

Place: Chennai
Date: September 2026

Signature of the Student
Ramya K
(2025503042)

ABSTRACT

The **DIGITAL COLLEGE LIBARY** is a web-based library management and book availability system designed to help students easily search, locate, and reserve books available in their college library.

In a traditional library, students may spend considerable time searching for a particular book and finding out whether it is available or already borrowed. Even when a book is available, students may have difficulty identifying its exact physical location.

The proposed system solves this problem by providing a centralized digital platform where students can search books using the book title, author, ISBN, category, or keywords. The system displays the total number of copies, available copies, borrowed copies, and reserved copies. Most importantly, every physical copy of an available book is displayed with its **exact floor, section, shelf, rack, and row location**.

Students can reserve an available physical copy through the website. Administrators or librarians can manage books, individual copies, categories, locations, users, reservations, and borrowing information.

The system provides a simple, user-friendly and efficient way to access library information and reduces the time required to locate books physically.

INTRODUCTION

Libraries contain a large number of books distributed across different sections, floors, shelves, and racks. Finding a particular book manually can be time-consuming, especially when students do not know the exact location of the book.

The **DIGITAL COLLEGE LIBARY** provides a digital solution for this problem. It allows students to search the library collection through a web interface and immediately obtain information about a book's availability and physical location.

For example, when a student searches for **"C++"**, the system can display all matching books along with the number of copies available and borrowed.

If a book has five physical copies, the system tracks each copy separately.

Example:

copy	**status**	**Location**
Copy 1	Available	Floor 1→shelf A12→Rack 03
Copy 2	Borrowed	---
Copy 3	Available	Floor 1→shelf A13→Rack 03
Copy 4	Available	Floor 1→shelf B05→Rack 02
Copy 5	Borrowed	---

The system therefore provides both **digital availability information and physical book-location information**.

PROBLEM STATEMENT

Traditional library systems may not provide students with sufficient information about the real-time availability and physical location of individual book copies.

Students may face the following problems:

1. Difficulty searching for books manually.
2. Lack of information about the number of available copies.
3. Difficulty identifying whether a book is borrowed or available.
4. No information about the exact shelf or rack location.
5. Students may waste time searching different floors and sections.
6. No convenient way to reserve an available book.
7. Library staff may have difficulty maintaining individual copy information.
8. Managing books, users, and reservations manually can result in errors.

Therefore, there is a need for a simple web-based system that provides book search, availability information, exact physical location, and reservation functionality.

OBJECTIVES

The main objectives of the DIGITAL COLLEGE LIBARY system are:

- To provide an easy-to-use online library search system.
- To allow students to search books quickly.
- To display the total number of book copies.
- To display available, borrowed, and reserved copies.
- To track each physical book copy independently.
- To provide the exact floor, section, shelf, rack, and row location.
- To allow students to reserve available books.
- To prevent duplicate reservations.
- To allow students to view and manage their reservations.
- To allow administrators to manage books and physical copies.
- To allow administrators to manage categories and library locations.
- To provide user and administrator management.
- To reduce the time required to locate books in the physical library.

PROPOSED SYSTEM

The proposed system is a web-based **DIGITAL COLLEGE LIBARY** that provides students with digital access to library book information.

The system consists of two main users:

Student

Students can:

- Register and log in.
- Search for books.
- Filter books by category.
- View book details.
- Check availability.
- View exact physical locations.
- Reserve available copies.
- View reservations.
- Cancel reservations.

Administrator / Librarian

Administrators can:

- Manage books.
- Manage physical copies.
- Manage categories.
- Manage locations.
- Manage students.
- Manage reservations.
- Manage borrowed and returned books.

- View library statistics.

The most important feature of the proposed system is **individual physical-copy tracking**.

Instead of treating five copies of a book as one record, the system maintains information about every physical copy separately.

MODULES

The DIGITAL COLLEGE LIBARY system consists of the following five major modules.

1. SEARCH & INFORMATION RETRIEVAL MODULE

The Search & Information Retrieval Module allows students to find books quickly.

Functions

- Search by book title.
- Search by author.
- Search by ISBN.
- Search by Book ID.
- Search by keywords.
- Filter by category.
- Filter by availability.
- Filter by floor.
- Sort search results.

Example

A student searches:

C++

The system displays:

- Book title
- Author

- Category
- Total copies
- Available copies
- Borrowed copies
- Reserved copies
- Exact location of available copies

This module reduces the time required to manually search the library.

2. CATEGORY MANAGEMENT MODULE

The Category Management Module organizes books into different categories.

Example categories:

- Computer Science
- Programming
- Engineering
- Mathematics
- Physics
- Chemistry
- Electronics
- Mechanical
- Civil
- Management
- General
- Reference

Admin Functions

The administrator can:

- Add category.
- Edit category.
- Delete category.
- View categories.
- Assign books to categories.

Students can use categories to filter search results.

3. INFORMATION DISPLAY MODULE

The Information Display Module presents complete information about books and their physical copies.

Book information

- Book title
- Author
- ISBN
- Publisher
- Edition
- Category
- Description
- Book cover
- Total copies
- Available copies
- Borrowed copies
- Reserved copies

Physical copy information

Each copy contains:

- Copy ID
- Status
- Building
- Floor
- Section
- Shelf
- Rack
- Row

Example:

C++ Programming – Copy 001

Status: Available

Location:

Main Library → Floor 2 → Computer Science Section → Shelf B12 → Rack 03 → Row 4

This allows students to physically locate the book.

4. CONTENT MANAGEMENT MODULE

The Content Management Module allows the administrator to maintain library information.

Book Management

Admin can:

- Add books.
- Edit books.

- Delete books.
- Add physical copies.
- Edit physical copies.
- Delete physical copies.
- Update book status.

Location Management

Admin can manage:

- Buildings
- Floors
- Sections
- Shelves
- Racks
- Rows

Borrowing Management

Admin can:

- Mark a copy as borrowed.
- Record borrower details.
- Set expected return date.
- Mark a copy as returned.
- Restore its available status.

5. USER & ADMIN MANAGEMENT MODULE

This module manages students and administrators.

Student Functions

- Registration
- Login
- Logout
- Profile management
- View reservations
- Cancel reservations

Admin Functions

- View students
- Search students
- Edit student information
- Enable/disable accounts
- Delete accounts
- View reservations
- Manage library information

The module also provides separate access for students and administrators.

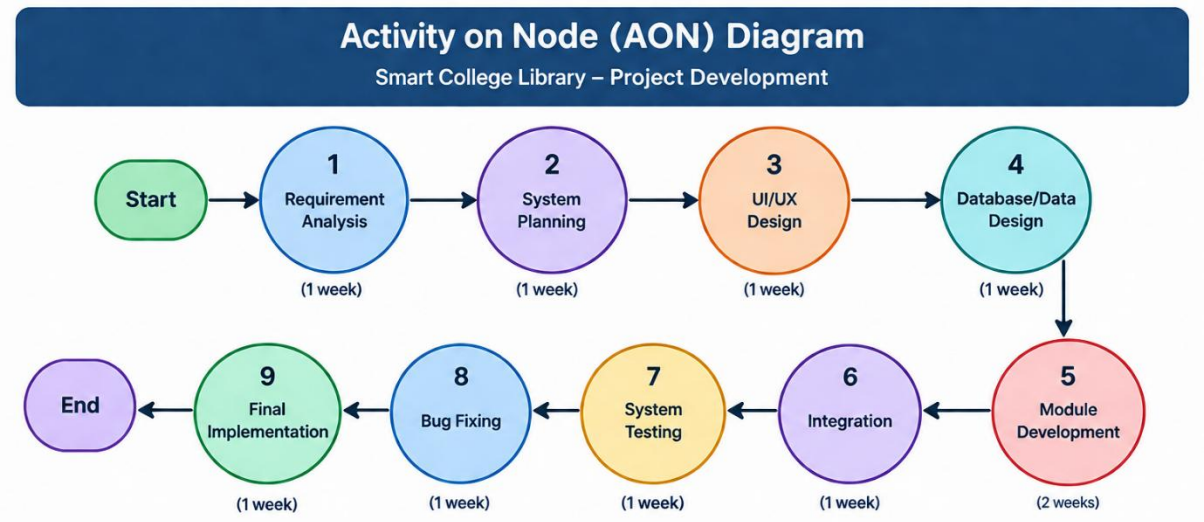
AON / AOE DIAGRAM

Project development can be represented using **Activity on Node (AON)** and **Activity on Edge (AOE)** diagrams.

ACTIVITY ON NODE (AON) DIAGRAM

In an AON diagram, activities are represented using nodes, while arrows represent the sequence or dependency between activities.

AON Flow



Activity No.	Activity Name	Duration	Dependencies
1	Requirement Analysis	1 week	-
2	System Planning	1 week	1
3	UI/UX Design	1 week	2
4	Database/Data Design	1 week	3
5	Module Development	2 weeks	4

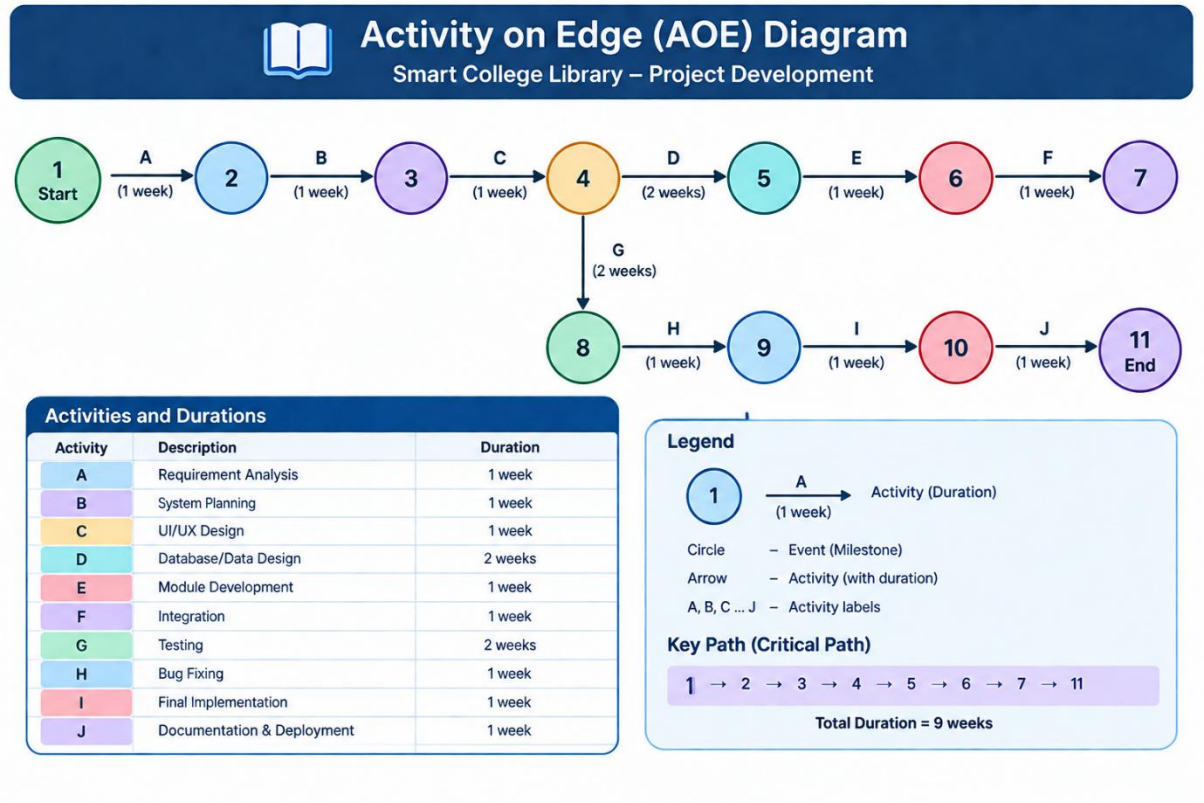
Activity No.	Activity Name	Duration	Dependencies
6	Integration	1 week	5
7	System Testing	1 week	6
8	Bug Fixing	1 week	7
9	Final Implementation	1 week	8

Activities

Activity	Description
Requirement Analysis	Identify library system requirements
System Planning	Plan system features and modules
UI/UX Design	Design pages and user interface
Data Design	Design books, copies, users and reservations
Module Development	Develop required modules
Integration	Connect all modules
Testing	Test system functionality
Bug Fixing	Correct identified errors
Implementation	Prepare final working system

7.2 ACTIVITY ON EDGE (AOE) DIAGRAM

In an AOE diagram, activities are represented by arrows and events are represented by nodes.



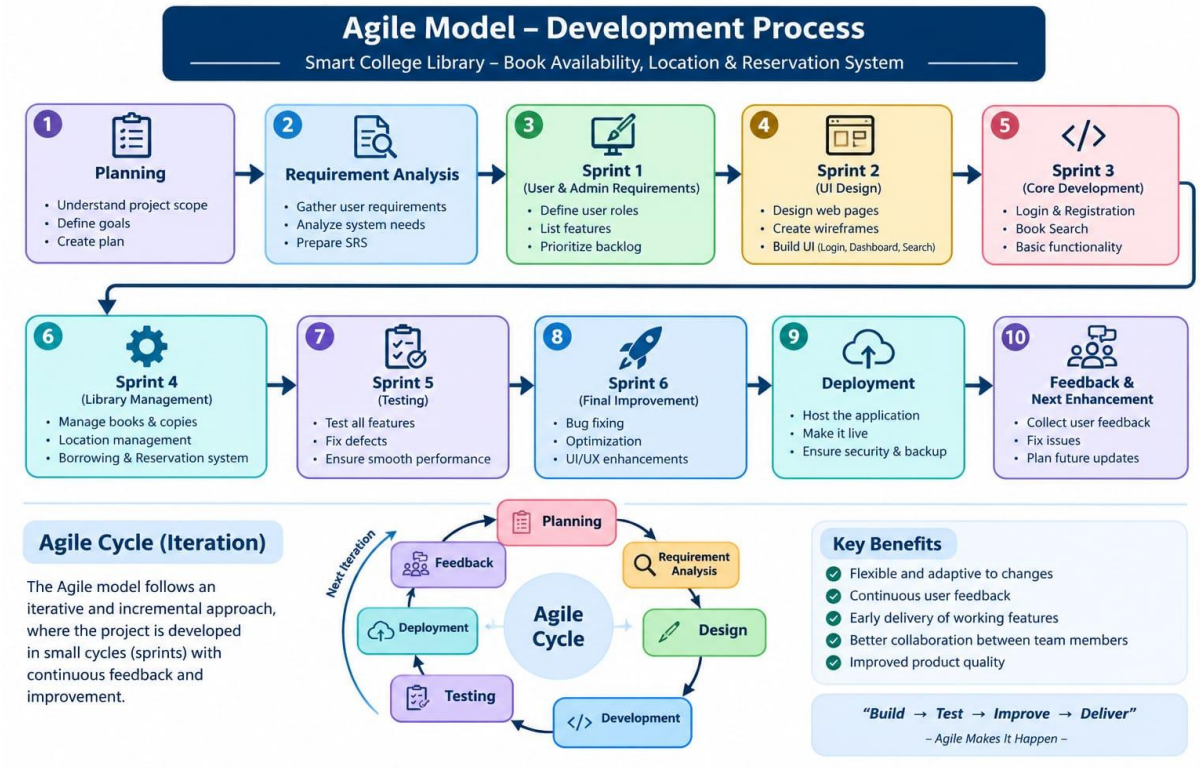
The AOE model shows the sequence of activities required to complete the project.

AGILE MODEL / SDLC

The **Agile Software Development Model** is used for developing the DIGITAL COLLEGE LIBARY system.

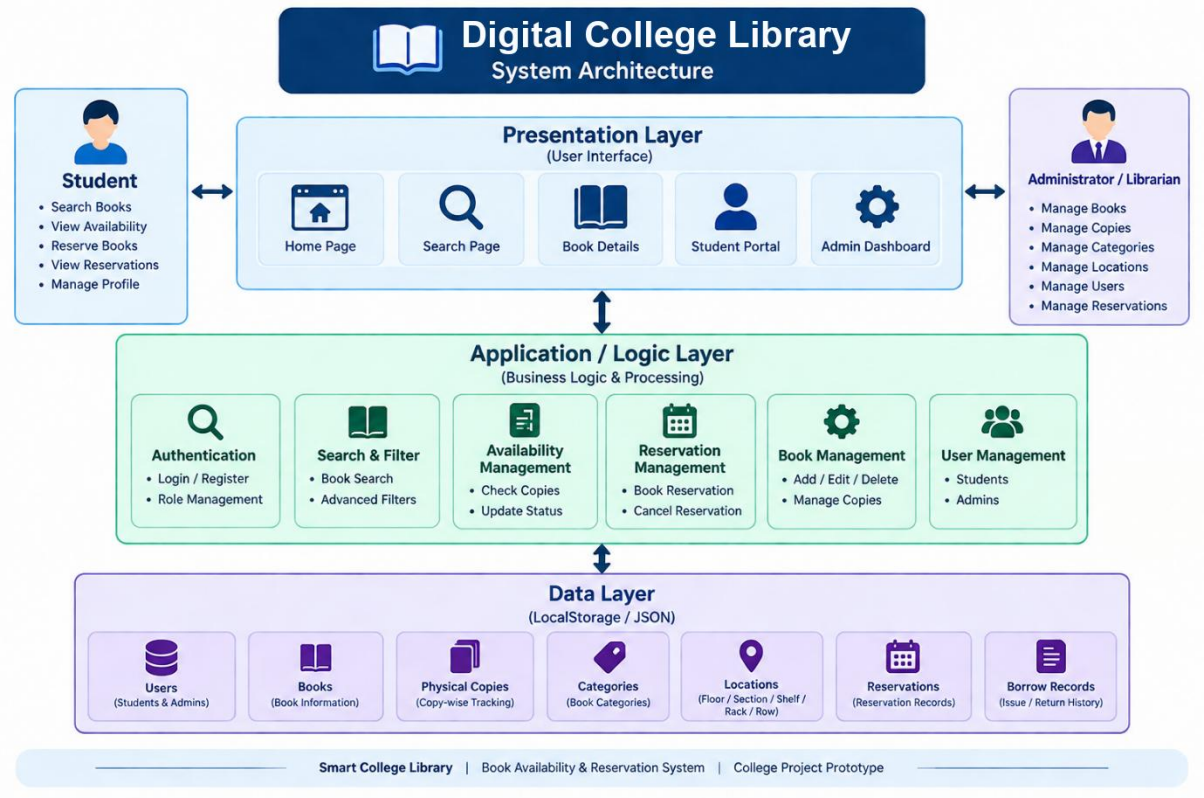
Agile development divides the project into smaller iterations. Each iteration includes planning, development, testing and improvement.

Agile Development Process



SYSTEM ARCHITECTURE

The DIGITAL COLLEGE LIBARY uses a simple layered architecture.



Presentation Layer

Provides the user interface for students and administrators.

Application Layer

Handles:

- Authentication
- Search
- Filtering

- Reservations
- Availability
- Book management
- User management

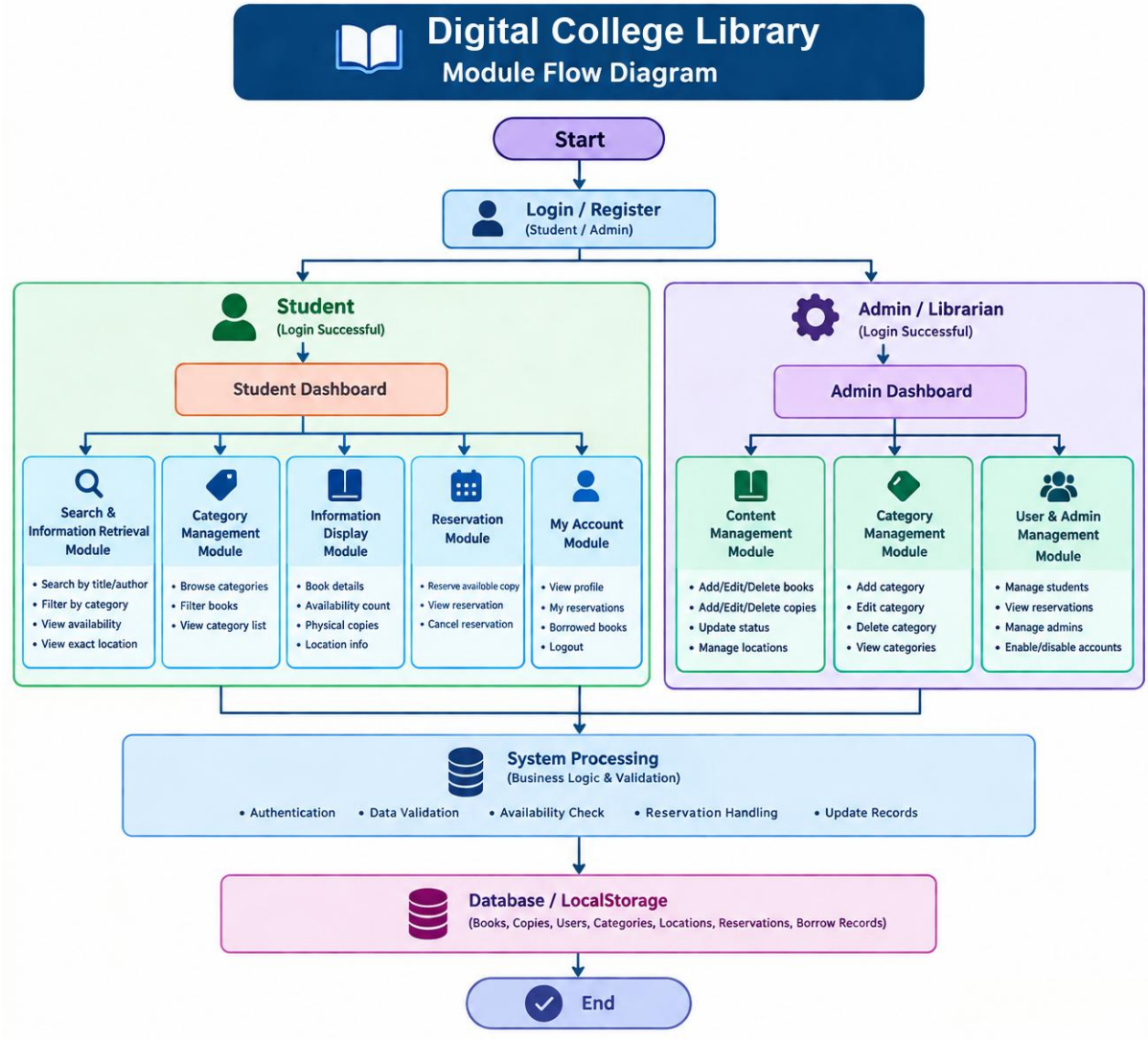
Data Layer

Stores information about:

- Users
- Books
- Copies
- Categories
- Locations
- Reservations
- Borrow records

MODULE FLOW DIAGRAM

The overall module flow is:



TECHNOLOGIES USED

The following technologies can be used to develop the DIGITAL COLLEGE LIBARY prototype.

HTML5

HTML5 is used to create the structure of the web pages.

Examples:

- Login forms
- Search interface
- Book cards
- Tables
- Navigation
- Admin dashboard

CSS3

CSS is used to design the user interface.

It provides:

- Layout
- Colors
- Typography
- Responsive design
- Buttons
- Cards

- Tables
- Animations

JavaScript

JavaScript is used to provide interactive functionality.

It handles:

- Search
- Filtering
- Login validation
- Registration
- Reservations
- Availability updates
- Dynamic book information
- Admin operations

LocalStorage / JSON

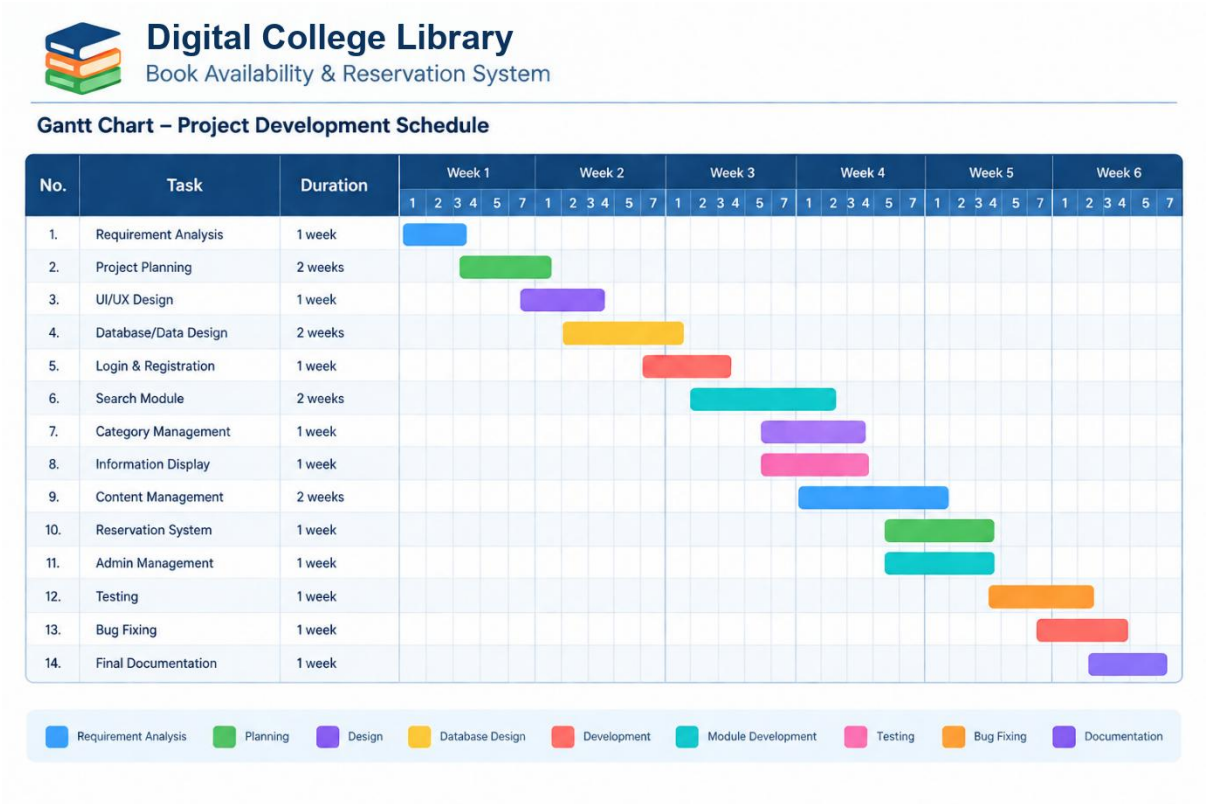
LocalStorage or JSON-based data can be used for the project prototype.

It stores:

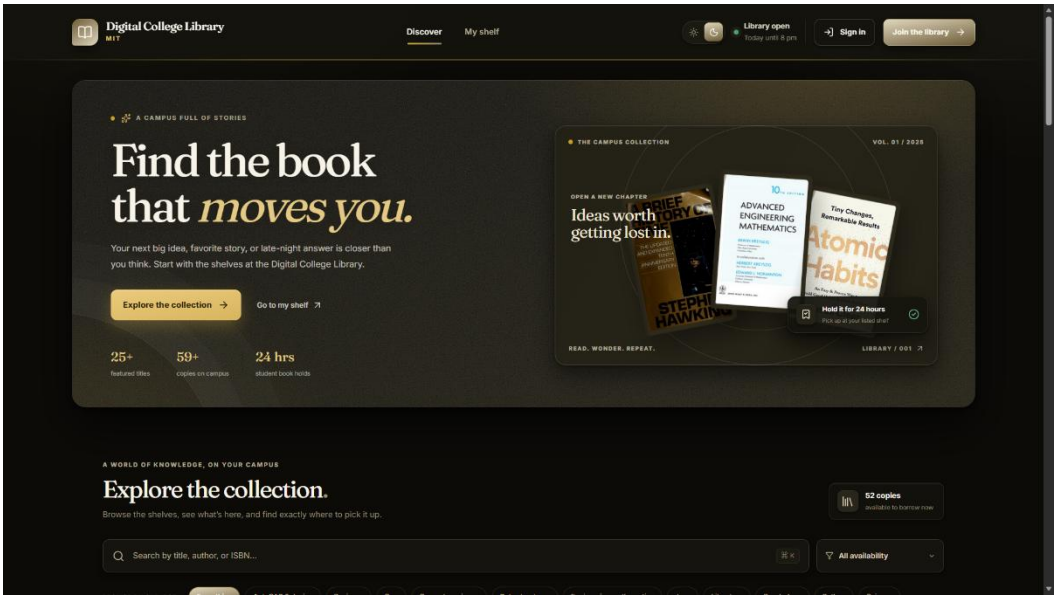
- User information
- Book information
- Physical copies
- Categories
- Reservations
- Borrowing records

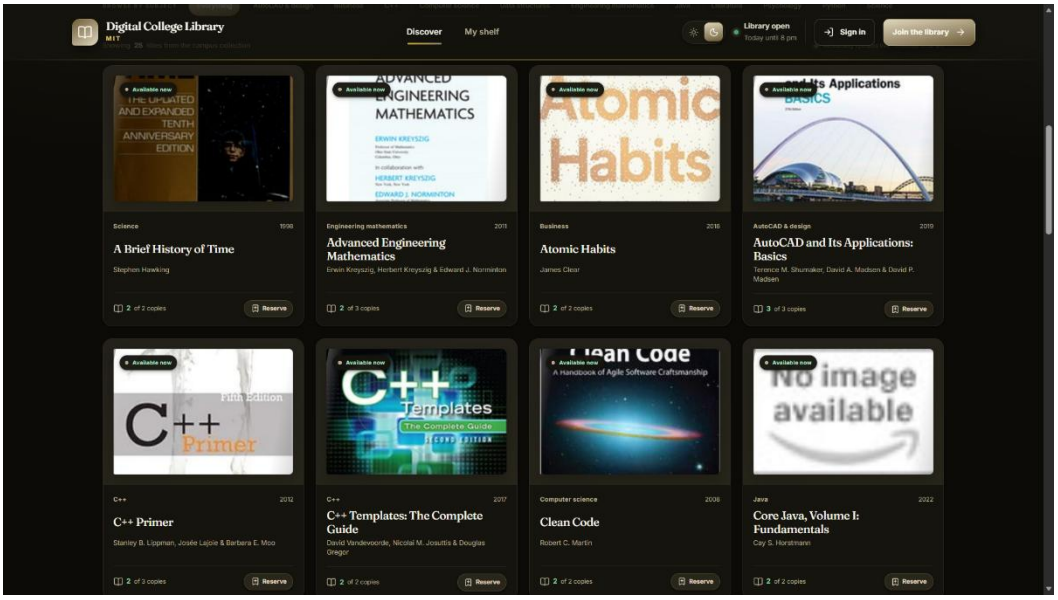
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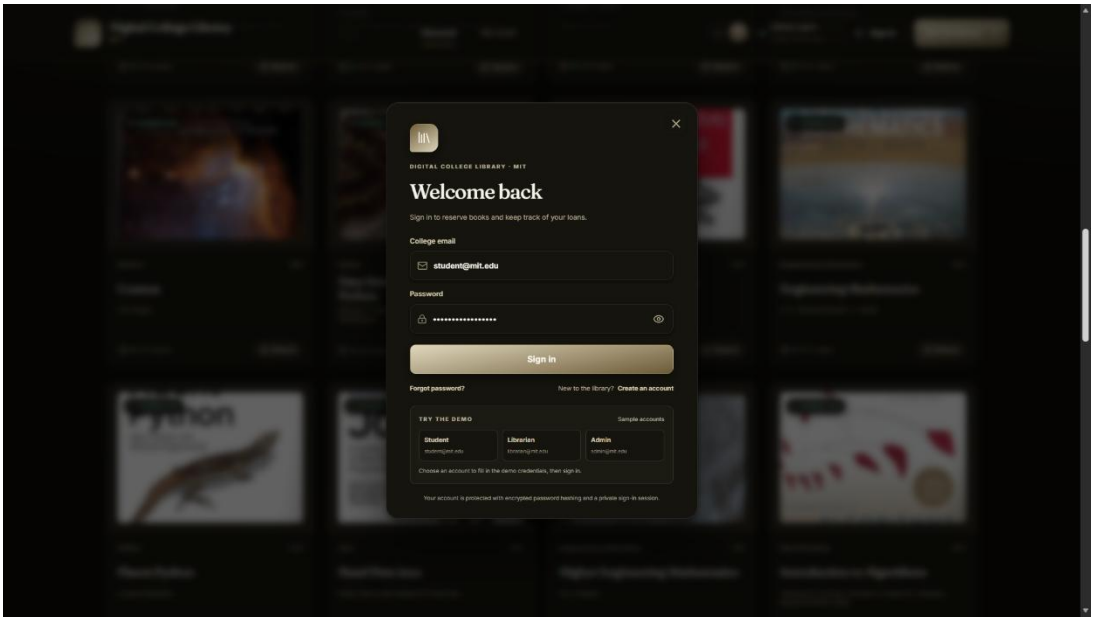
The following is a sample project development schedule.



SCREENSHOTS







TESTING

Testing is performed to verify that the DIGITAL COLLEGE LIBARY system works correctly.

Functional Testing

Test ID	Test Case	Expected Result	Status
TC01	Student registration	Account created successfully	Pass
TC02	Student login	Student dashboard opens	Pass
TC03	Invalid login	Error message displayed	Pass
TC04	Search for C++	Matching books displayed	Pass
TC05	Search by author	Matching books displayed	Pass
TC06	Search by ISBN	Correct book displayed	Pass
TC07	Category filter	Books filtered correctly	Pass
TC08	View book details	Complete book information displayed	Pass
TC09	View available copies	All available copies displayed	Pass
TC10	View physical location	Floor/shelf/rack displayed	Pass
TC11	Reserve available copy	Reservation created	Pass
TC12	Duplicate reservation	Duplicate reservation prevented	Pass
TC13	Cancel reservation	Copy becomes available	Pass
TC14	Admin login	Admin dashboard opens	Pass
TC15	Add book	Book added successfully	Pass
TC16	Edit book	Book information updated	Pass
TC17	Delete book	Book removed successfully	Pass
TC18	Add physical copy	Copy added successfully	Pass
TC19	Change copy status	Status updated correctly	Pass
TC20	User management	Admin can manage users	Pass

Usability Testing

The interface is tested to ensure that:

- Navigation is simple.
- Search is easy to find.
- Book information is clearly displayed.
- Availability status is understandable.
- Physical location is easy to identify.
- Reservation process is simple.
- Admin functions are organized.

Responsive Testing

The website should be tested on:

- Desktop
- Laptop
- Tablet
- Mobile

The layout should automatically adjust according to screen size.

FUTURE ENHANCEMENTS

The DIGITAL COLLEGE LIBARY system can be further improved with the following features:

1. Real Database

Replace LocalStorage/JSON with:

- MySQL
- PostgreSQL
- MongoDB

This would allow multiple users to access the same real-time library database.

2. QR Code Integration

Each physical book copy can have a QR code.

Scanning the QR code could display:

- Book information
- Copy ID
- Location
- Current status

3. Barcode Scanner

Barcode scanning can be used by librarians for:

- Book issue
- Book return
- Copy identification

4. Email Notifications

Students can receive notifications about:

- Reservation confirmation
- Reservation expiry
- Book return dates
- Overdue books

5. Online Fine Management

The system can calculate overdue fines automatically.

6. Library Floor Map

An interactive floor map can highlight the exact shelf where the book is located.

7. AI-Based Book Recommendations

The system could recommend books based on:

- Previous searches
- Borrowing history
- Department
- Interests

8. Mobile Application

The system can be converted into an Android/iOS application.

9. Real-Time Availability

A centralized database can provide real-time availability information across multiple library branches.

CONCLUSION

The **DIGITAL COLLEGE LIBARY – Book Availability, Location & Reservation System** provides an efficient digital solution for searching and locating books in a college library.

The system allows students to search for books, view complete book information, check the number of available and borrowed copies, identify the exact physical location of every available copy, and reserve books conveniently.

The system also provides administrators with tools to manage books, physical copies, categories, library locations, users, reservations, and borrowing information.

The key advantage of the system is its **individual physical-copy tracking**. Instead of simply showing that a book is available, the system identifies exactly where each available copy is located, such as the floor, section, shelf, rack, and row.

The project demonstrates how a web-based information system can improve the accessibility, organization, and efficiency of a college library while reducing the time students spend searching for books.

Therefore, the DIGITAL COLLEGE LIBARY provides a simple, user-friendly and practical solution for modern college library management.

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